

Mini Data Platform in Docker Containers – Semester Project Guide

Objective

Develop a mini data platform using Docker containers that simulates a business process, ingests data into PostgreSQL, captures changes with Debezium, streams data through Kafka, processes it in Spark and stores it in MinIO in Delta format.

Task-by-Task Breakdown & Deliverables

Task 1: Simulating a Business Process with Python

Topics

- Writing a Python script to simulate a business process.
- Reading data from three CSV files.
- Creating tables in PostgreSQL dynamically.
- Inserting data into PostgreSQL.

Deliverables

- Python script that:
 - Reads three CSV files.
 - Creates PostgreSQL tables dynamically.
 - Inserts data into PostgreSQL.
 - Updated `docker-compose.yml` with PostgreSQL
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Task 2: Connecting Debezium to Capture Changes in PostgreSQL

Topics

- Setting up Debezium to monitor PostgreSQL changes.
- Configuring Debezium as a Kafka connector.
- Enabling JSON serialization for messages sent to Kafka.

Deliverables

- Configured Debezium service in `docker-compose.yml`.
- PostgreSQL configured with logical replication enabled.
- Debezium configured to use JSON format for messages.
- Verified that changes in PostgreSQL are captured by Debezium and sent to Kafka in JSON format.

Task 3: Kafka Setup & Streaming Events in AVRO Format

Topics

- Setting up a Kafka cluster in Docker.
- Configuring Kafka topics to use JSON format.
- Setting up Schema Registry for JSON message serialization.
- Writing a Kafka consumer to validate incoming messages.

Deliverables

- Running Kafka cluster in `docker-compose.yml`.
- Kafka configured with JSON serialization.
- Schema Registry set up to manage JSON schemas.
- Kafka consumer script that reads JSON messages and decodes them.

Task 4: Integrating Spark with Kafka for Data Processing

Topics

- Introduction to Apache Spark and its role in big data processing.
- Connecting Spark Structured Streaming to Kafka.
- Deserializing JSON messages from Kafka in Spark.
- Performing basic transformations on streamed data.

Deliverables

- Spark container added to `docker-compose.yml`.
- Spark job that reads JSON messages from Kafka.
- Basic transformation performed on incoming data.

Task 5: Storing Processed Data in MinIO using Delta Lake

Topics

- Introduction to MinIO as an S3-compatible object store.
- Writing Spark DataFrames in Delta format.
- Configuring MinIO for Spark access.

Deliverables

- MinIO configured in `docker-compose.yml`.
- Spark job that writes processed data to MinIO in Delta format.

- Documentation on how to retrieve data from MinIO.
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Task 6: Automating Deployment & Ensuring Reliability

Topics

- Automating the deployment process.
- Using Docker volumes and networks for better integration.
- Handling failures and ensuring data consistency.

Deliverables

- Fully automated docker-compose.yml with all services.
- Updated documentation on how to deploy the project from scratch.